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New MovMie:

Leveraging Collaborative Filtering and a Multilayer Perceptron Network for Movie Recommendations

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Introduction

Indecisive people know the questions: “Where should we go to eat?”, “What book should I read next?”, “What should we watch?”. They grapple with the desire to try something new, only to settle on The Office for the tenth time in a row because nothing else looks appealing. Other recommendation systems, such as TikTok or YouTube, are designed to align as perfectly as possible with a user’s interests based on their history. This algorithmically creates an “echo chamber,” where the user receives similar content over and over again. It enables apps to become addictive, leading to doom-scrolling, impacted mental health, and affected work efficiency (Hou et al., 2019).

We initially tested using collaborative filtering (CF) alone and a multilayer perceptron (MLP) alone to generate a suggestion for what to watch, but these results were geared towards making the best possible recommendation with the given information, which is what the other recommendation systems currently do. Further, simply flipping the results around (i.e. recommending the worst possible movie) would not have taken into account the nuances of wanting a recommendation that the user will like and rate highly, but is unlikely to try on their own.

Leveraging a combination of user-user CF and a MLP, New MovMie is a recommendation system intended to encourage people to break out of their chambers, watch something new, and gain perspective. New MovMie filters CF data by user and MLP data by unrated movies by the user, it pinpoints CF predicted movie ratings that are higher than MLP predicted movie ratings, and it identifies the movies associated with the highest CF predicted ratings as recommendations. This integration uniquely uses the results from both methods to develop an unlikely new recommendation.

Literature Review

The process of using a MLP algorithm to predict movie ratings based on a variety of input features is well documented in the literature. One study in particular used a meta-path-based heterogeneous information network (HIN) embedding model to extract nodes’ structural features; these features represented information about users and items based on the network’s structure and different

types of relationships. Then, the extracted structural features were converted and fused using a nonlinear method. Finally, the fused features were provided to the MLP as input, and the MLP predicted the rating a user would give to a specific item. The performance of the MLP was evaluated on movie rating datasets like Movielens-100k and Douban Movie to demonstrate the effectiveness of the approach (Yi et al. 2021, 728-36).

Collaborative filtering has also been applied to ranking videos. Zhao et al. (2019) explores the challenges within designing large-scale multi-objective ranking systems specifically for video sharing through a two-stage design: candidate generation and ranking. The paper touches on the “candidate generation” step of the design, which is used to select the potentially relevant subset of data to then use for the recommendation (Zhao et al. 2019). In Zhao et al. 's (2019) model, the ranking system uses deep neural networks to make the ultimate recommendation after candidate generation, so they used filtering to identify relevant users. Ultimately, Zhao et al. (2019) found that while CF-based systems generate candidates efficiently, they fall short in recommendations compared to neural networks.

Conversely, Mohamad et al. (2022) relied on a combination of item-item collaborative filtering and content-based filtering to develop their movie recommendation system to incorporate shared preferences among users. They explore model-based versus memory-based methods, where the former creates a model to predict future user behavior, and the latter leverages nearest neighbors to assume similar users have similar preferences (Mohamad et al. 2022). Content-based filtering takes in user preferences in one of three filtering types: 1) wrapper methods, using the subset of features with the most potential, 2) filter methods, using heuristic methods, and 3) embedded algorithms, choosing features during training (Mohamad et al. 2022). The combination of content-based and item-item collaborative filtering also took form in three types of hybrid systems: 1) weighted, where context sources are given weighted scores based on user history, 2) mixed hybrid, where the top *k* from each list of context source rankings are selected, and 3) cross-source hybrid, identifying popular items across sources to recommend (Mohamad et al. 2022). Mohamad et al. (2022) ultimately suggest the combination of content-based and

collaborative filtering to address the limitations of the approaches on their own in the context of recommendation systems.

Machine Learning Methodology and Data Collection

Originally, the data were meant to be scraped from a popular social media platform like Instagram or TikTok, but given time constraints, the data were changed to the Movielens-100k dataset. The data were collected by downloading the ml-100k.zip file from the GroupLens website; extracting the README, u.data (full dataset), u.item (information about the items), and u.user (information about the users) files; and uploading the files to a GitHub repo.

Multilayer Perceptron

Specific to the MLP, data curation began by reading in the u.data, u.item, and u.user files as Pandas DataFrames; merging u.data and u.item on the item ID; and merging the merged DataFrame and u.user on the user ID. Second, the user ID and item ID columns were dropped from the final merged DataFrame. Third, “release_date”, “video_release_date”, and “IMDb_url” were dropped as features because each of those features had missing values. Fourth, an additional intercept/bias column was added to the data. Fifth, all categorical features were one-hot encoded. Sixth, the resulting DataFrame was turned into a NumPy Array for ML purposes. Seventh, Min-Max Scaling was performed on all features except the movie rating. Lastly, the data was separated into input features and target.

Once the data was curated, the implementation of the MLP began by setting a random seed to maintain the same values in the weight matrices. Second, the features, target, step size of 0.1, and two weight matrices with respective dimensions of 2500x10 and 10x1 (e.g. w^(2) and W^(1)) were instantiated. Third, a helper function was written that defined the ReLU activation function for the hidden layer and the sigmoid activation function for the output, the latter being appropriate for MLP for classification. Fourth, an epoch size of 500 was instantiated; for each epoch, the derivative with respect to w^(2) was calculated, w^(2) was updated using the old value of w^(2) and the current values of the hidden layer, the derivative with respect to W^(1) was calculated, W^(1) was updated, and the gradient

descent error was calculated using the log loss objective function, which is also appropriate for MLP for classification. The only difference from course content was this MLP was trained on all the data.

Collaborative Filtering

The methodology of the CF model was performed based on user similarities. First, the u.user file was read into R as a table of three columns: UserID, MovieID, and the rating. Next, the data were reshaped with the users as the indices and the movies as the columns, populated by ratings. By user (row), data is centered by subtracting the mean of the user's ratings from each rating. Then, missing values were imputed at the mean, 0. The similarity score was calculated between each user, for both the lsa library's cosine and the L2 norm, then min/max scaled. The *k* most similar users determined via similarity score were evaluated, with four different *k*-values tested: 3, 5, 10, and 20. For each (cosine, k) pair, we calculated the rating by weighting similar users' ratings for the item by the similarity scores and dividing by the sum of the similarity scores, following the below calculation.

$$r(u_i, I_j) = \frac{r(u_a, I_j) * Sc(u_i, u_a) + r(u_b, I_j) * Sc(u_i, u_b)}{Sc(u_i, u_a) + Sc(u_i, u_b)}$$

This particular aspect of the project did not vary significantly from the collaborative filtering performed in the course. However, the incorporation of testing the algorithm via masking out a "test" set allowed us to more effectively measure the efficacy of collaborative filtering and delve a bit deeper than we were able to in class. Given the slow nature of the model, we limited the testing set to the first 10 users in the masked set: 196, 186, 22, 244, 166, 298, 115, 253, 305, and 6.

Hybrid Model

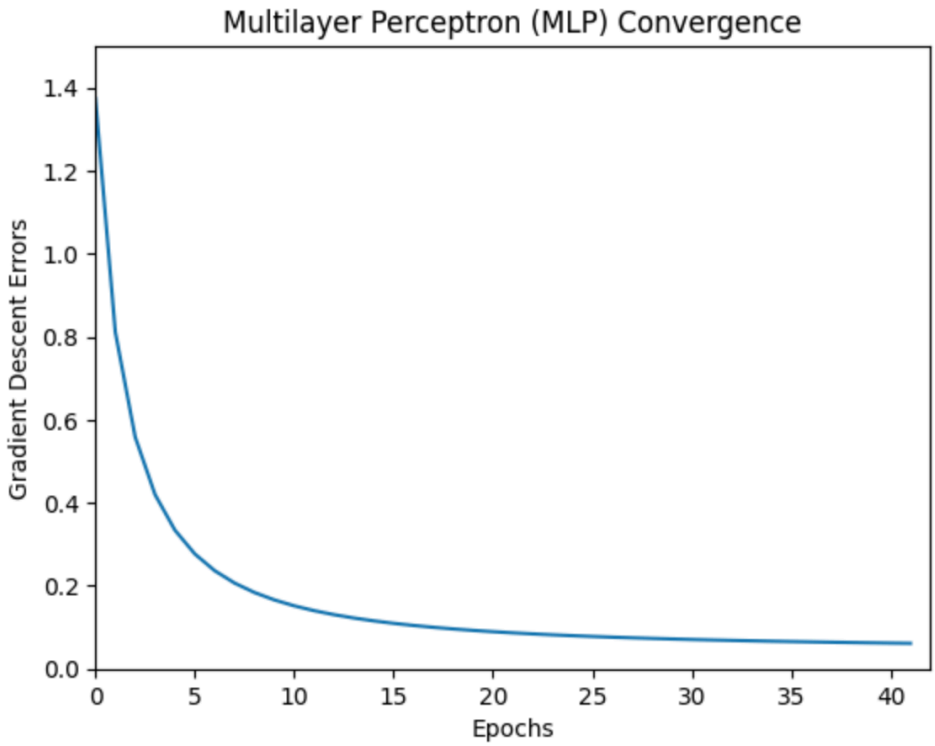
This project explored the combination of MLP-informed CF decision making to make final decisions. First, the MLP and CF prediction datasets were read in as DataFrames. Second, the CF prediction data was filtered for each user in the masked set, and the filtered data was sorted by movie ID in ascending order. Third, the MLP prediction data was filtered for the same unrated movies in the CF prediction data for each user and sorted by movie ID in ascending order. Fourth, the CF and MLP DataFrames for each user were merged on movie ID, rows of the merged DataFrame were filtered for

higher CF predicted ratings than MLP predicted ratings, the filtered DataFrame was sorted by CF predicted rating in descending order, and the highest rating was identified, serving as a recommendation.

Machine Learning Results and Interpretations

Multilayer Perceptron

After the MLP algorithm ran for 500 epochs, the final weight estimates for W^(1) and w^(2) as well as the predicted output were displayed as a sanity check, and the gradient descent errors were plotted as a function of the number of epochs, as seen in Img 1. One thing to note is that, although the algorithm was run for 500 epochs, error values were only produced in 42 epochs because the error calculation involved division by zero and other improper calculations for certain epochs. In addition to those results, the ratings were predicted using the MLPClassifier() function from scikit-learn, the MLP's accuracy was evaluated at approximately 0.480, the MLPClassifier() predictions were joined to the user and movie IDs from the original data, and a .csv file of the results was created to be used for further analysis.



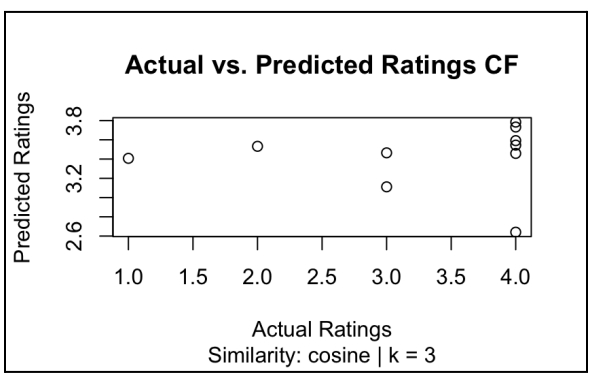
Img 1: Gradient Descent Errors vs. Epochs for MLP Convergence

Collaborative Filtering

Similarity Metric	k-value	Root Mean Square Error	Mean Absolute Error	Rounded Accuracy
cosine	3	**1.049144**	0.7764208	**0.6**
cosine	5	1.115880	0.7928518	0.5
cosine	10	1.088645	0.8268106	0.4
cosine	20	1.080632	0.8113918	**0.6**
L2	3	1.232339	0.8748830	0.5
L2	5	1.212395	0.8458676	0.5
L2	10	1.258693	0.8074147	**0.6**
L2	20	1.199227	**0.7278715**	**0.6**

Table 1: Collaborative Filtering Fine-Tuning

Both the cosine and L2 similarity calculations for four different k-values: 3, 5, 10, and 20 were tested. For each similarity calculation and k-value, metrics collected included root mean square error (RMSE), mean absolute error (MAE), and accuracy. Given the nature of the CF model to return predictions as doubles, we rounded the values to the nearest whole number for the accuracy metric only. Looking for the lowest RMSE and highest rounded accuracy, we found cosine, k=3 to be the best set of hyperparameters. While this model was outperformed by L2, k=20 for MAE, the difference of about 0.05 was deemed more acceptable than the RMSE difference of ~0.15. Thus, we relied on cosine with a k-value of 3 in the generation of our predictions.



Img 2: Actual vs. Predicted Ratings Scatterplot for Collaborative Filtering, cosine & k=3

Img. 2 displays the extracted test users and their CF-predicted ratings plotted against their actual ratings for the top performing parameters of the model. The majority of actual ratings were 4.0, which also had the largest variation in prediction. Lower ratings were sparser and overestimated.

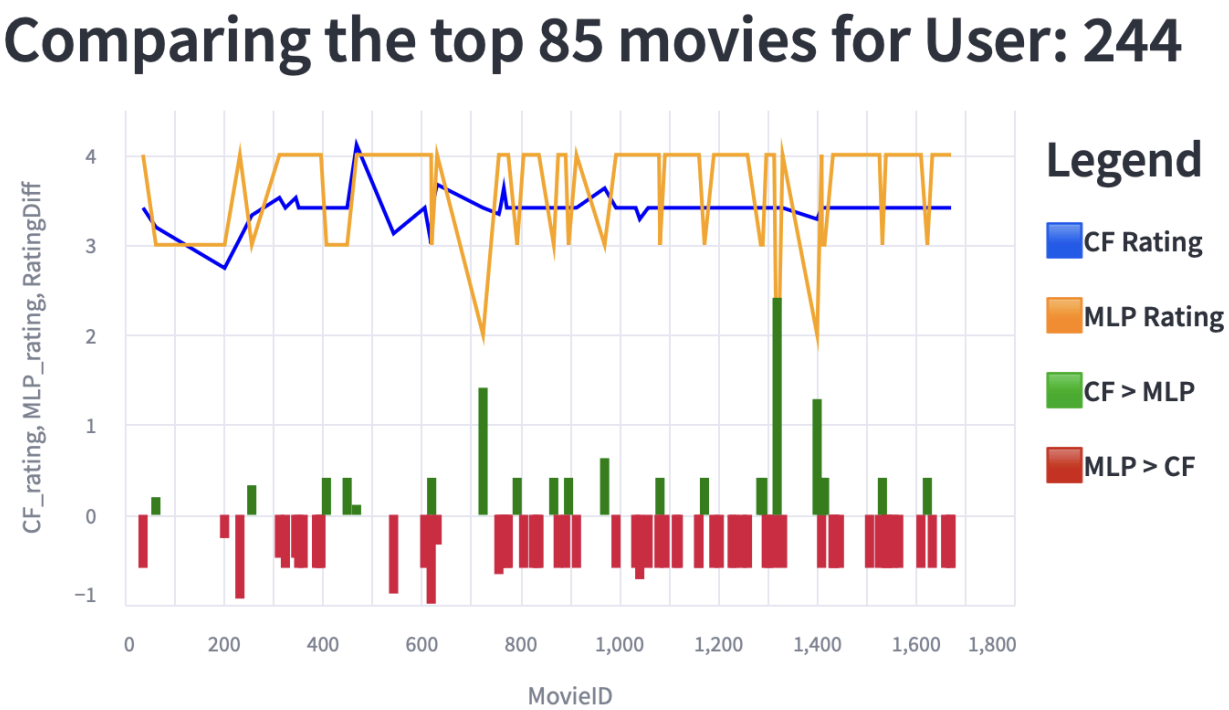
Hybrid Model

Once the highest rating and the associated movie ID was identified for each user, the movie ID was cross-referenced with the merged dataset from the MLP algorithm to identify the genre(s) of the recommendation. The merged dataset was then filtered for each user ID, and three movies rated as 5 were chosen and their genres were identified as preferred genres of the user. The CF-MLP merged DataFrame for each user was then used to extract the CF and MLP ratings for the recommendation.

Starting with User 196, a few movies this user rated as a 5 include Movie 655 (Stand by Me (1986)), Movie 67 (Ace Ventura: Pet Detective (1994)), and Movie 663 (Being There (1979)); given these movies, User 196 seems to like Comedies. The hybrid model found that Movie 357 (One Flew Over the Cuckoo's Nest (1975)), which falls under the Drama genre, was the unrated movie for this user with the highest CF rating of about 4.25, compared to a MLP rating of 4.0. Thus, Movie 357 serves as a proper recommendation for User 196.

User 186 rated Movie 566 (Clear and Present Danger (1994)), Movie 470 (Tombstone (1993)), and Movie 925 (Unforgettable (1996)) as a 5; this says that User 186 seems to like attention-grabbing movies. The hybrid model found that Movie 286 (The English Patient (1996)), which falls under the Drama/Romance/War genres, was the unrated movie with the highest CF rating of about 4.61, compared to a MLP rating of 4.0. Thus, Movie 286 serves as a proper recommendation for User 186.

User 22 rated Movie 128 (Supercop (1992)), Movie 258 (Contact (1997)), and Movie 510 (The Magnificent Seven (1954)) as a 5; this says that User 22 seems to like Action movies. The hybrid model found that Movie 183 (Alien (1979)), which falls under the Action/Horror/Sci-Fi/Thriller genres, was the unrated movie with the highest CF rating of about 4.33, compared to a MLP rating of 4.0. Thus, Movie 183 serves as a proper recommendation for User 22.



Img 3: Visualization of All Rated CF vs. MLP Ratings for User 244 - extracted from NewMovMie_App.py

User 244 rated Movie 154 (Monty Python's Life of Brian (1979)), Movie 89 (Blade Runner (1982)), and Movie 652 (Rosencrantz and Guildenstern Are Dead (1990)) as a 5; this says that User 244 seems to like Comedies. The hybrid model found that Movie 483 (Casablanca (1942)), which falls under the Drama/Romance/War genres, was the unrated movie with the highest CF rating of about 4.67, compared to a MLP rating of 4.0. Thus, Movie 483 serves as a proper recommendation for User 244.

User 166 rated Movie 328 (Conspiracy Theory (1997)), Movie 322 (Murder at 1600 (1997)), and Movie 300 (Air Force One (1997)) as a 5; this says that User 166 seems to like Thrillers. The hybrid model found that Movie 228 (Star Trek: The Wrath of Khan (1982)), which falls under the Action/Adventure/Sci-Fi genres, was the unrated movie with the highest CF rating of about 3.90, compared to a MLP rating of 3.0. Thus, Movie 228 serves as a proper recommendation for User 166.

User 298 rated Movie 8 (Babe (1995)), Movie 498 (The African Queen (1951)), and Movie 496 (It's a Wonderful Life (1946)) as a 5; this says that User 298 seems to like feel-good movies. The hybrid model found that Movie 192 (Raging Bull (1980)), which falls under the Drama genre, was the unrated

movie with the highest CF rating of about 4.45, compared to a MLP rating of 4.0. Thus, Movie 192 serves as a proper recommendation for User 298.

User 115 rated Movie 8 (Babe (1995)), Movie 127 (The Godfather (1972)), and Movie 234 (Jaws (1975)) as a 5; this says that User 115 seems to like attention-grabbing movies. The hybrid model found that Movie 169 (The Wrong Trousers (1993)), which falls under the Animation/Comedy genres, was the unrated movie with the highest CF rating of 5.0, compared to a MLP rating of 4.0, Thus, Movie 169 serves as a proper recommendation for User 115.

User 253 rated Movie 465 (The Jungle Book (1994)), Movie 510 (The Magnificent Seven (1954)), and Movie 183 (Alien (1979)) as a 5; this says that User 253 seems to like attention-grabbing movies. The hybrid model found that Movie 195 (The Terminator (1984)), which falls under the Action/Sci-Fi/Thriller genres, was the unrated movie with the highest CF rating of about 4.13, compared to a MLP rating of 4.0. Thus, Movie 195 serves as a proper recommendation for User 253.

User 305 rated Movie 427 (To Kill a Mockingbird (1962)), Movie 483 (Casablanca (1942)), and Movie 50 (Star Wars (1977)) as a 5; this says that User 305 seems to like Dramas. The hybrid model found that Movie 23 (Taxi Driver (1976)), which falls under the Drama/Thriller genres, was the unrated movie with the highest CF rating of about 4.36, compared to a MLP rating of 4.0. Thus, Movie 23 serves as a proper recommendation for User 305.

Lastly, User 6 rated Movie 14 (Il Postino (1994)), Movie 98 (The Silence of the Lambs (1991)), and Movie 492 (East of Eden (1955)) as a 5; this says that User 6 seems to like Dramas. The hybrid model found that Movie 654 (Chinatown (1974)), which falls under the Film-Noir/Mystery/Thriller genres, was the unrated movie with the highest CF rating of about 4.67, compared to a MLP rating of 4.0. Thus, Movie 654 serves as a proper recommendation for User 6.

Conclusions and Future Work

The MLP is based on a few assumptions, including those about the complexity of the objective function; the assumption that we have the input, hidden layer(s), and output readily available and that we

just need to find the weights; and the assumption that we'll get the same solution each time since a seed is set to initialize the weights. Overall, the assumptions were met and the analysis of the data was appropriate, but certain choices that were made may have inhibited the performance of the algorithm. To elaborate, if we were to continue working on the problem, some things we could do include implement a train-test split on the data instead of train on the full data, change the step size of gradient descent, change the number of nodes in the hidden layer and/or the number of hidden layers, change the activation functions for the hidden layer and the output layer, and change the number of epochs to run the algorithm for.

The CF is based on the assumption that people who had similar preferences in the past will continue to have such in the future. Also, we assumed that the rest of the data would follow a similar pattern to the small sample we picked. Overall, the assumptions were met and the analysis of the data was appropriate, but if we were to continue working on the problem, we'd do more cross validation, use bigger samples, and/or test more users. In addition, the data was skewed in that far more people rated medium-high (3.0-5.0, 4.0-5.0) than low, particularly within our test set which were overwhelmingly rated at 4.0. This may have created an overabundance of higher predicted ratings for users. In general, the skew of the predictions towards higher than or around the actual values is reflective of the tendency of users to recommend things they greatly enjoyed, rather than those that were "OK" (~3.0) or "bad" (<3.0).

At large, this project was limited in its data. We relied on the 100K MovieLens Dataset, which was released in April 1998—about 27 years prior to our applications of CF and MLP. Conducting a similar project on a more recent dataset is a promising future endeavour, especially as the number of tracked metrics has likely increased.

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